LINEAR ALGEBRA · LESSON 8

Solving Ax = b

Can a client's target be hit? Which portfolios hit it? **Four short steps** on one matrix.

None, one, or infinitely many answers.

[R | d] $x = x_p + x_n$ MIT 18.06, Lecture 8

THE QUESTION

Hit ₹10L, ₹105K, exposure 7

Funds: Nifty, Nifty ETF, bonds, balanced.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 10 \\ 12 & 12 & 7 & 9.5 & 105 \\ 1 & 1 & 0 & 0.5 & 7 \end{array}\right]$$

1. Simplify to $[R \mid d]$
2. Check the zero rows
3. One answer: free amounts = 0
4. Every answer: add do-nothing trades

STEP 1

Simplify to [R | d]

Same five row operations as Lesson 7, also on the last column:

$$[\,R \mid d\,] = \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0.5 & 7 \\ 0 & 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Last column: $105 - 120 = -15,\ 7 - 10 = -3,\ -3 + 3 = 0,$ $-15/-5 = 3,\ 10 - 3 = 7.$

STEP 2

Check the zero rows

Row 3 reads $0 = d_3$. No trade can make 0 equal anything else.

Target	Row 3 ends in	
(10, 105, 7)	0	possible ✓
(10, 105, 5)	−2	impossible ✗

Exposure is already decided by budget and return for these funds: asking for 5 is asking for two things at once.

STEP 3

One answer: read it off

Put 0 in the free funds (ETF, balanced). Each row of $[R \mid d]$ gives a pivot amount:

$$\text{Nifty} = 7, \quad \text{bonds} = 3$$

ONE PORTFOLIO

$x_p = (7, 0, 3, 0)$: ₹7L Nifty, ₹3L bonds. Budget 10 ✓, return $84 + 21 = 105$ ✓, exposure 7 ✓.

STEP 4

Every answer

Pick any free amounts, read the rest. ETF = 3, balanced = 2: bonds $= 3 - 1 = 2$, Nifty $= 7 - 3 - 1 = 3$.

$$x = \underbrace{(7, 0, 3, 0)}_{x_p} + \underbrace{c(-1, 1, 0, 0) + d(-0.5, 0, -0.5, 1)}_{\text{do-nothing trades}}$$

One answer + any do-nothing trade = every answer.

HOW MANY ANSWERS?

Three rules

Look at $[R \mid d]$ for two things: a **contradiction** (a zero row like $0 = -2$) and **free variables.**

[R | d] shows	Answer	Example
contradiction	❌ none	twins, (10, 105): 0 = −15
no contradiction, no free	✅ one	Nifty + bonds: (7, 3)
no contradiction, free	∞ infinite	twins, (10, 120)

MEMORY TRICK

Contradiction → NONE. No contradiction + no free → ONE. No contradiction + free → INFINITE.

THE RANK DECIDES

Four cases

Rank	Answers	Funds
r = m = n	exactly 1	Nifty, bonds, midcap
r = n < m	0 or 1	Nifty, bonds, 3 rules
r = m < n	∞, always	Nifty, bonds, midcap, balanced
r < m, r < n	0 or ∞	Nifty, ETF, bonds, balanced

$r = m$: no zero rows, always possible. $r = n$: no free columns, at most one.

IN MARKETS

Every way to build a payoff

Futures, 20,000 call, 20,000 put, 19,900 call: rank 3 = 3 scenarios, so every payoff is buildable. For $(-200, 100, 900)$:

$$x = (1, 2, 0, 1) + t(1, -1, 1, 0)$$

"1 fut + 2 calls + 1 × 19,900 C" = "3 fut + 2 puts + 1 × 19,900 C". Quote whichever is cheaper.

RECAP

The four steps

1. Simplify $[A \mid b]$ to $[R \mid d]$.
2. Contradiction (0 = non-zero) ⇒ no answer.
3. Free amounts 0 ⇒ pivot amounts $= d$.
4. Every answer = that one + any do-nothing trade.
5. Contradiction → NONE; no free → ONE; free → INFINITE.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A target tester, a lab that solves any system up to 3 × 5 step by step, and worked hedging and options examples:

learn.krisarj.com

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